

Seminar Presentation

Contrastive Learning for Sequential Recommendation

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Xu Xie et al. (2022). "Contrastive learning for sequential recommendation". In: *2022 IEEE 38th international conference on data engineering (ICDE)*. IEEE, pp. 1259–1273

Introduction – Sequential Recommendation

- ▶ **Recommender Systems**

- Help users find relevant items (e.g., products, movies, news).
- Many approaches assume static user preferences.

- ▶ **Sequential Recommendation**

- Recommendations based on a user's interaction history.
- Able to better capture dynamic user interests over time.

Introduction – Sequential Recommendation

- ▶ Formulation as a Prediction Task
 - Given a historical interaction sequence of a user.
 - Predict the next item the user will interact with.
 - Very similar to the language modelling task in NLP.

Introduction – Challenges in Sequential Recommendation

► Data Sparsity

- Most users and items have only a few interactions.
- Difficult to learn robust user preference patterns from short sequences.

► Learning Effective User Representations

- Deep models have many parameters, need lots of data.
- Next-item prediction alone can overfit and struggle to generalize.

Introduction – Contrastive Learning

► Self-Supervised Learning

- Learning from the data itself without human-created labels.
- Define some pretext task based on the raw data.
- Has been effective in other domain like CV and NLP.

► Contrastive Learning

- The pretext task is to compare different instances.
- The model learns to tell similar things apart from dissimilar things.
- Positive pair: Two augmented versions from same input.
- Negative pair: Augmented versions of different inputs.

CL4SRec – The Proposed Framework

- ▶ User Representation Encoder

- Takes a user's interaction sequence as input.
- Produce a user representation as output.
- Can be implemented using different models.

- ▶ Encoder Shared Between Two Learning Objectives

- Recommendation Loss: Standard task of predicting next item in the sequence.
- Contrastive Loss: Compare augmented views from the original interaction sequences.

CL4SRec – Data Augmentation Strategies

▶ Item Crop → Randomly select a continuous subsequence of the sequence.

▶ Item Mask → Randomly replace some items with a special mask item.

▶ Item Reorder → Randomly shuffle items within a continuous subsequence.

CL4SRec – Model Architecture

- ▶ User Representation Model based on Transformer

- Items and position embedding layer.
- Multi-head self-attention
- Position-wise feed-forward network
- Layer normalization and dropout for regularization

- ▶ Next Item Probability Prediction

- Based on dot product between user and item representation.
- The items closest to the user representation are predicted.

CL4SRec – Training Process

► Recommendation Loss

- Typically cross-entropy, predicting the true next item versus other items.
- Uses only a sampled softmax for efficiency.
- $\mathcal{L}_{\text{main}}(s_u) = -\log \frac{\exp(s_u^\top v^+)}{\exp(s_u^\top v^+) + \sum_{v^- \in V^-} \exp(s_u^\top v^-)}$

► Contrastive Loss

- Aims to maximize similarity for positive pairs and minimize for negative pairs.
- $\mathcal{L}_{\text{cl}}(s_u^{a_i}, s_u^{a_j}) = -\log \frac{\exp(s_u^{a_i \top} s_u^{a_j})}{\exp(s_u^{a_i \top} s_u^{a_j}) + \sum_{s^- \in S^-} \exp(s_u^{a_i \top} s^-)}$

► Multi-Task Learning

- The model is trained to optimize both tasks simultaneously.
- $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{cl}}$

Evaluation — Experimental Setup

- ▶ Four public real-world datasets.

- Amazon Beauty, Amazon Sports, Yelp, MovieLens-1M.
- Filtered to only users and items with at least 5 interactions.
- Only uses item identifiers, not ratings or review text.
- Interactions grouped by user and ordered by timestamp.
- Last item for testing, Second-to-last item for validation.

- ▶ Evaluation Metrics

- HR@k: Ratio the correct item appears in the top-k recommendations.
- NDCG@k: Consider also the rank in the top-k at which the correct item appear.

Evaluation – Comparison With Other Methods

- ▶ Baselines for Comparison

- Non-personalized: Popularity-based
- Sequential: GRU4Rec+, SASRec

- Non-sequential: BPR-MF, NCF
- Other: GC-SAN, S³-Rec_MIP

- ▶ CL4SRec outperforms all baselines across all metrics and datasets.
 - Particularly effective on sparser datasets.

Evaluation – Influence of Hyper-Parameters

- ▶ → All three augmentation methods improve performance over not using CL.
- The best augmentation method varies by dataset.
- The proportion of augmentation is also an important hyperparameter.
- A well-chosen λ is crucial.

Evaluation – Data Augmentation vs. Contrastive Learning

- ▶ Full CL4SRec further improves over just applying augmentations.
 - Data Augmentation alone helps over standard SASRec.
 - Contrastive Learning on top helps more.

Evaluation – Better User Representations

- ▶ Experiment on Yelp dataset, which contains social graph of friends.
 - Compute the histogram of similarities between user representation of friends.
 - Friends have more similar representations under CL4SRec than under SASRec.
 - Suggests CL4SRec learns user representations that are “more meaningful”.

Evaluation – Strengths and Weaknesses of the Paper

► Strengths

- First to apply contrastive learning to sequential recommendation in this manner.
- Improves over state-of-the-art recommendation systems.
- Performs extensive evaluation to verify those claims.

► Weaknesses

- No theoretical analysis of why contrastive learning works well in this setting.
- They claim CL4SRec is “model-agnostic”, but only evaluated with a transformer model.
- Only compare to baseline models with differently tuned hyperparameters per dataset.

Quiz

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